Assignment No. 8

Title: Google App Engine.

Problem Statement: Installation and configure Google App Engine.

Objective: Installation and configure Google App Engine.

Outcome: Installation and configure Google App Engine.

Theory:

The Google App Engine (GAE) is a Platform as a Service (PaaS) offering from Google Cloud Platform (GCP). In a PaaS model, developers focus only on writing and deploying application code, while the cloud provider manages infrastructure such as virtual machines, operating systems, scaling, load balancing, and security updates.

App Engine provides two environments: Standard and Flexible.

In this experiment, the Standard environment was used, which runs applications in a secure sandbox and automatically scales based on traffic.

The developer does not need to configure servers manually.

Unlike Infrastructure as a Service (IaaS) platforms such as AWS EC2, where the user must install software, configure web servers, and manage ports, App Engine abstracts all infrastructure layers. The application is deployed using the gcloud app deploy command, and Google automatically provisions the required runtime environment.

Each App Engine project has:

• A project ID

• A region (selected during initialization)

• A default service URL (project-id.region.r.appspot.com)

The application configuration is defined using an app.yaml file. This file specifies the runtime environment (e.g., Python 3.11) and the entry point command used to run the application.

Dependencies are defined in a requirements.txt file.

During deployment, Google Cloud:

1. Uploads the source code.

2. Installs dependencies.

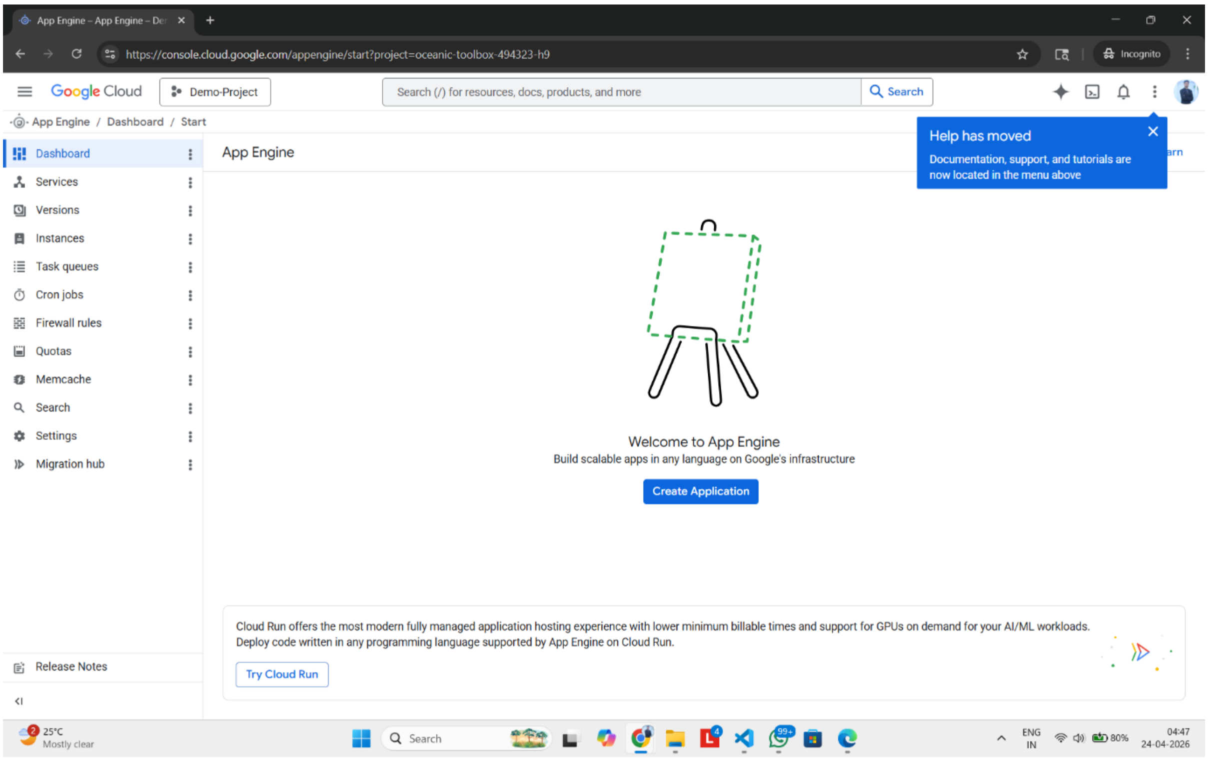
3. Creates a containerized runtime environment.

4. Starts the application using the specified entry point.

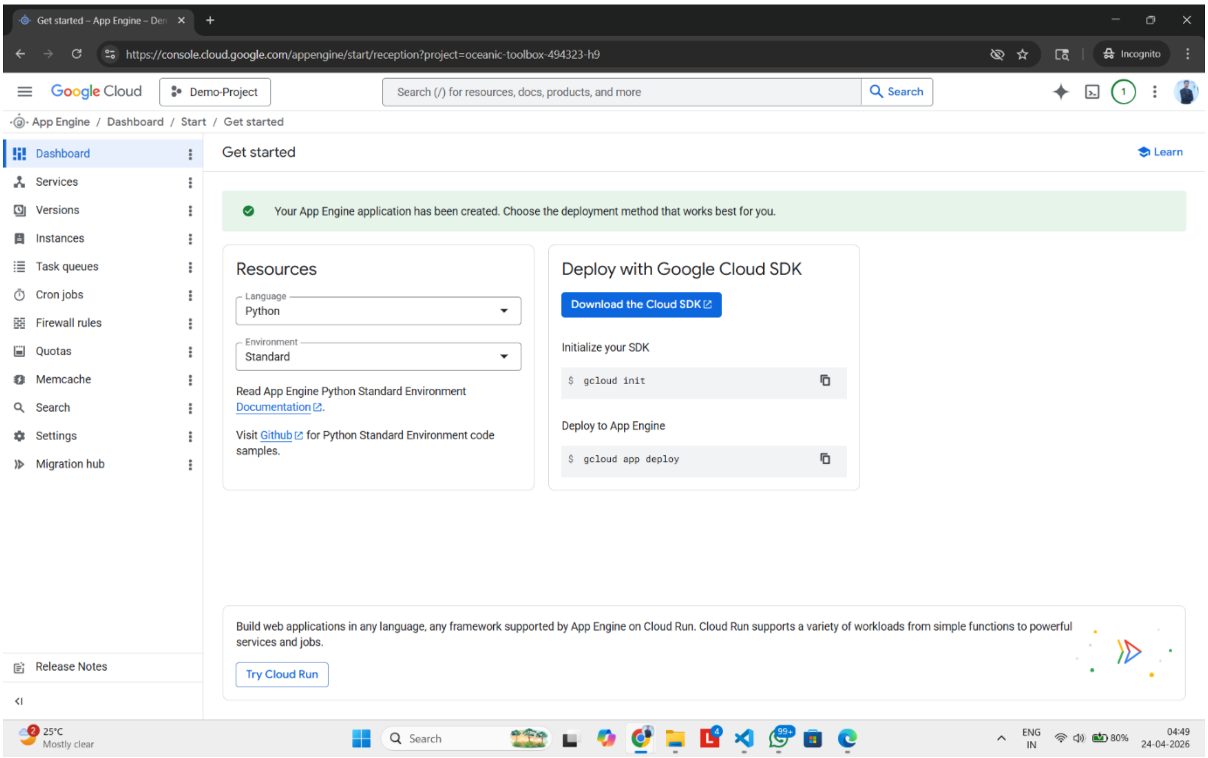
5. Assigns a public URL.

Procedure:

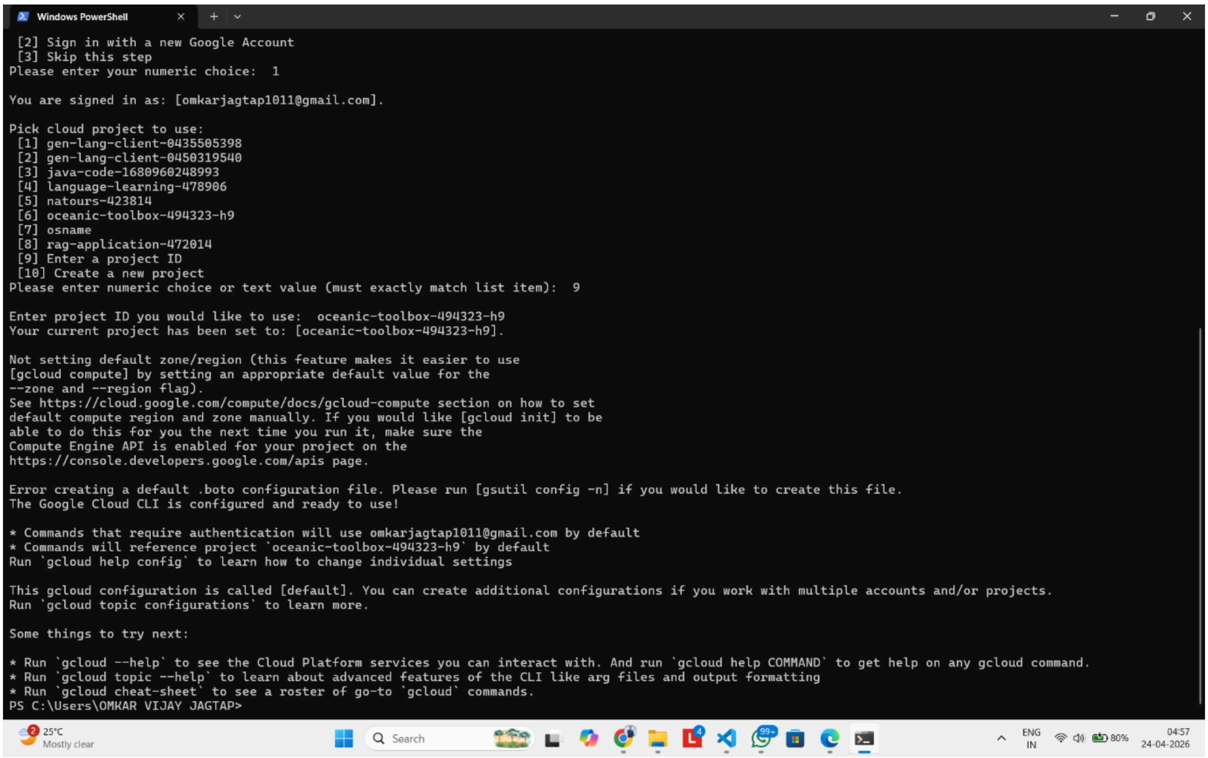
Step 1: Go to Google Console and Search App Enigne



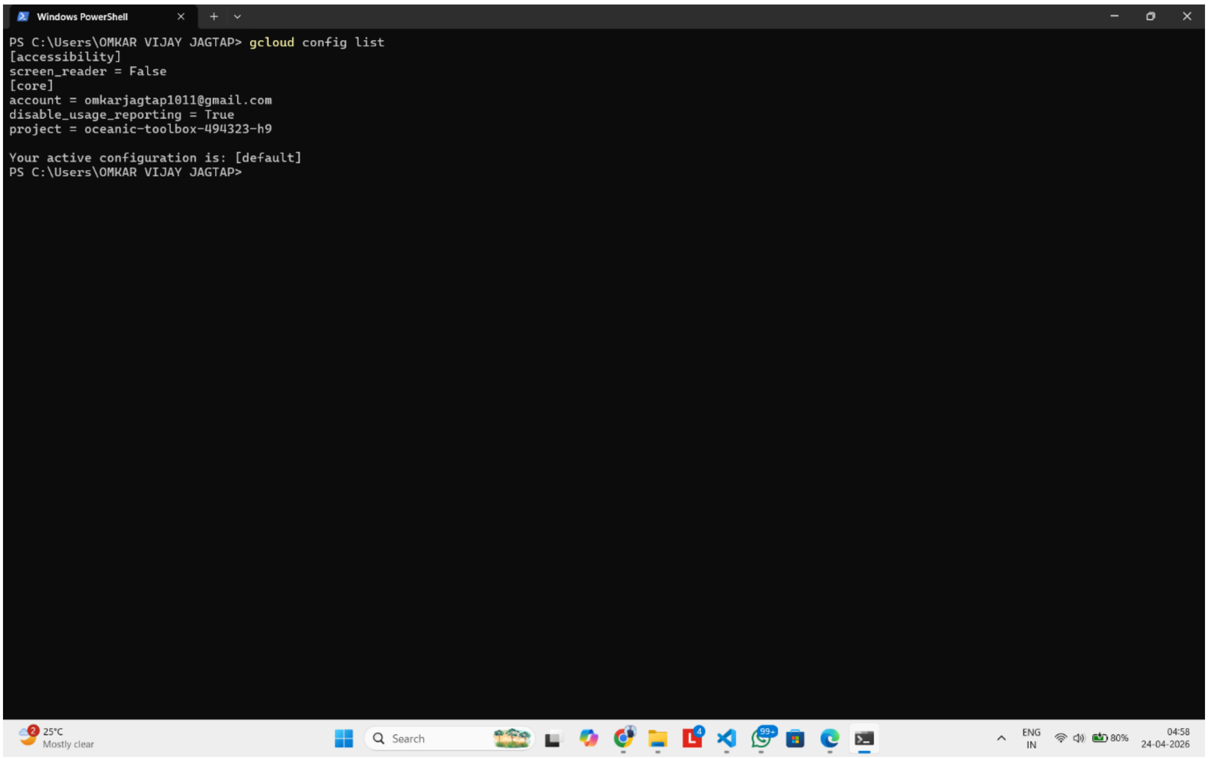
Step 2: Select the region and resources



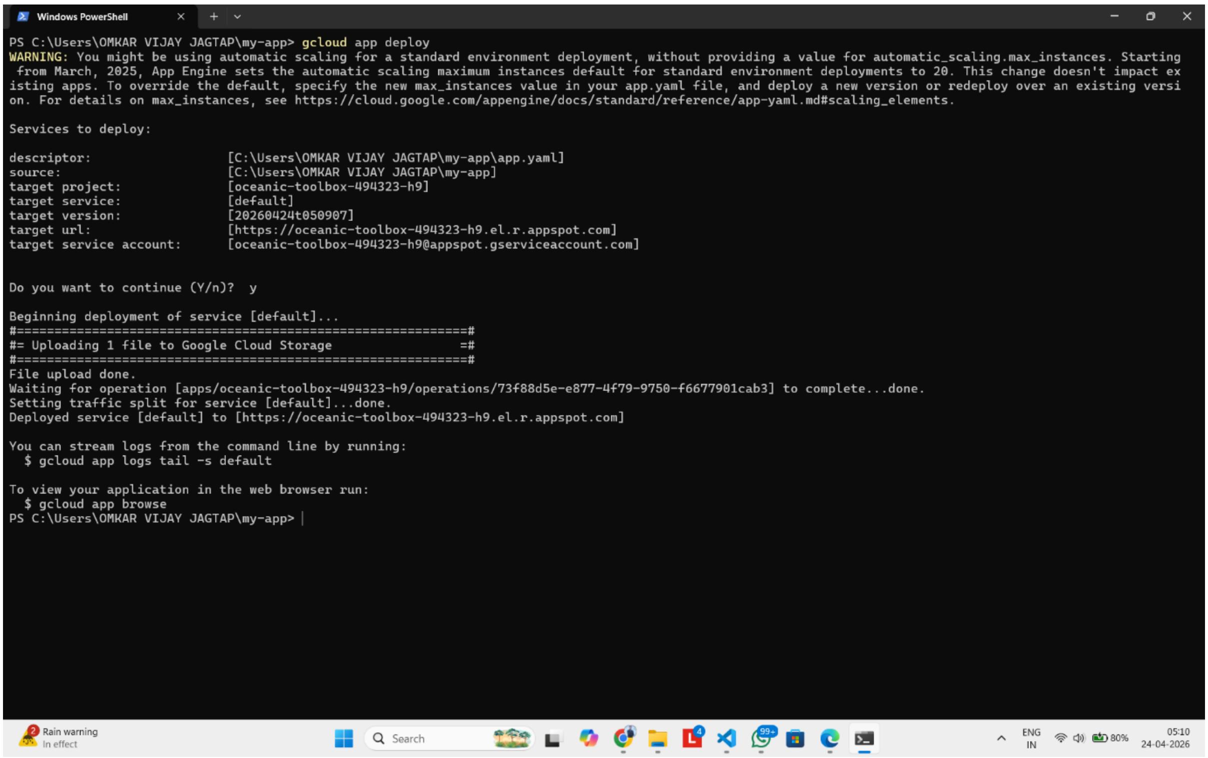
Step 3: Come to terminal and enter ‘gcloud init’



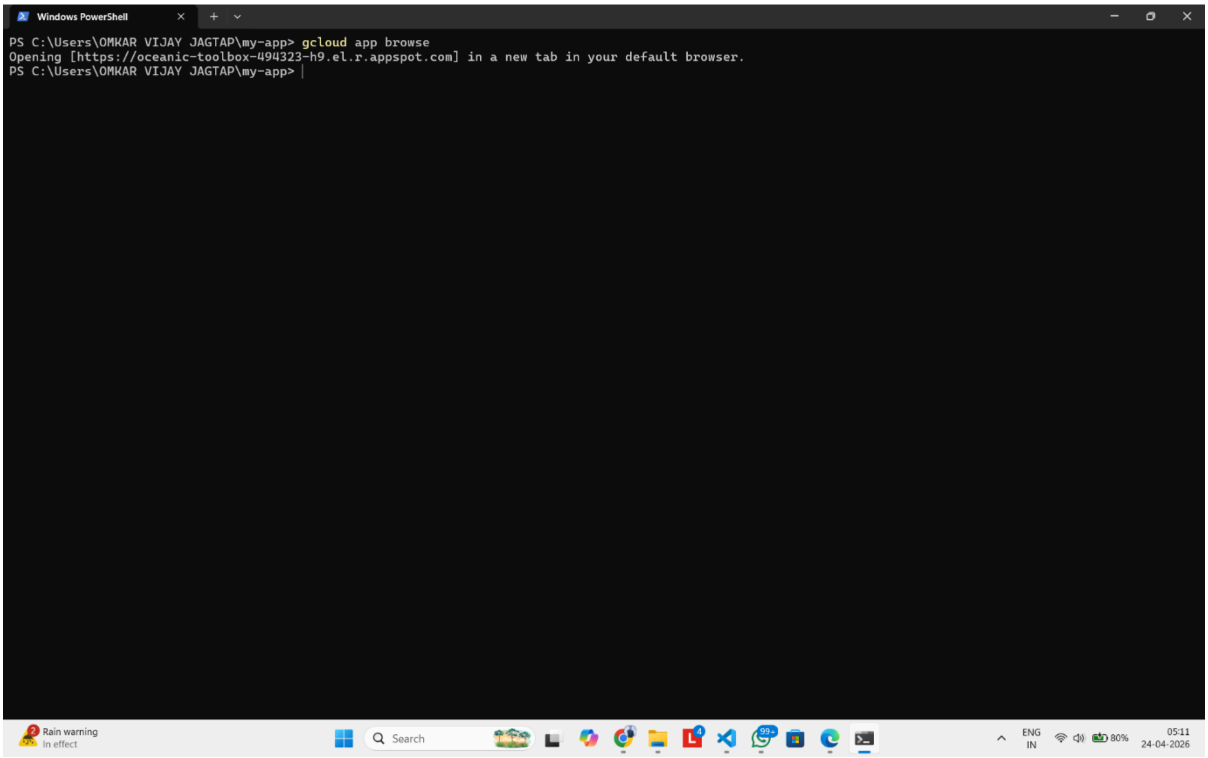
Step 4: Verify Config using ‘gcloud config list’

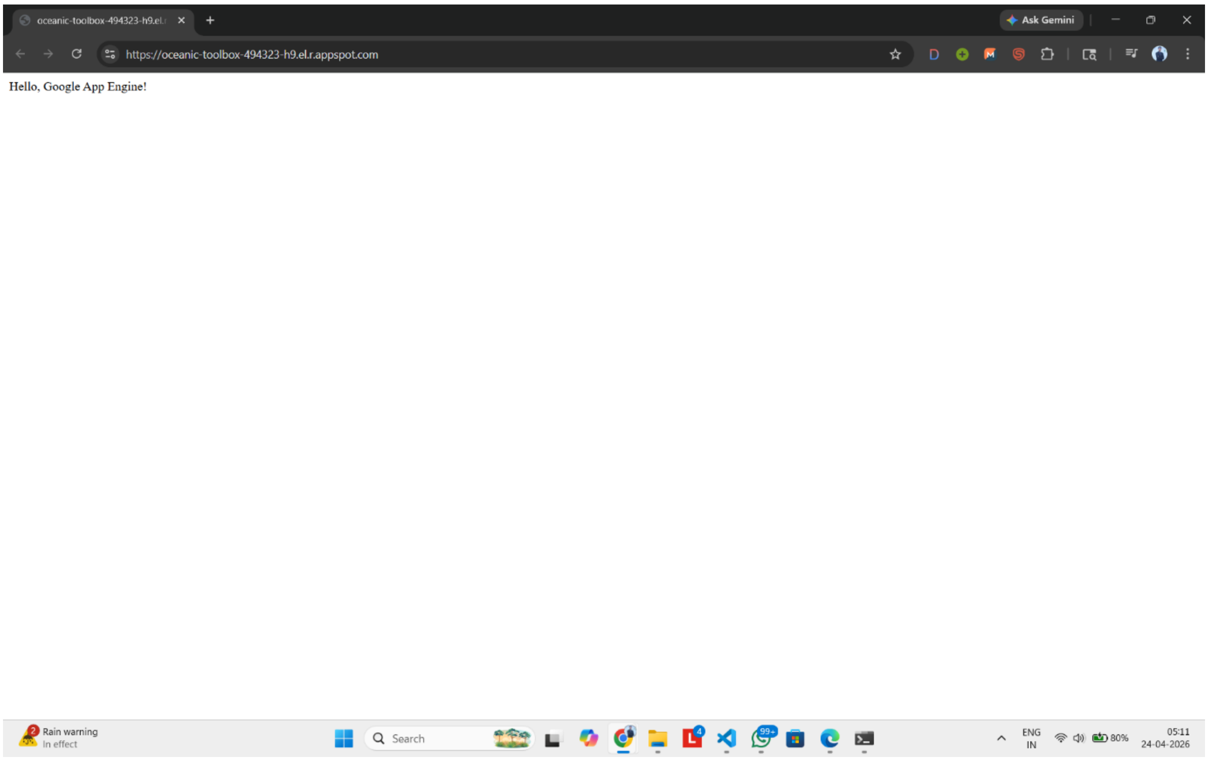


Step 5: Deploy the app using ‘gcloud app deploy’



Step 6: Launch app using ‘gcloud app browse’





Conclusion: Google App Engine was successfully installed and configured. A simple web application was deployed using the Python runtime. The experiment demonstrated the functioning of the Platform as a Service model and highlighted the difference between PaaS and IaaS cloud services